AI Ethics, Future of Work and Societal Impact

Lesson 1 to Lesson 6

Student Version — Original Booklet Extract (Reorganised)

Source: Education Bureau, Module on Artificial Intelligence
for Junior Secondary Level

This booklet reorganises six original units into Lesson 1 to Lesson 6.
The original content and page numbers have not been changed.
Divider pages are for lesson separation only.

Contents

Lesson	Topic	Original source	Pages in this booklet
Lesson 1	AI Ethical Principles	Booklet 1, Unit 3	Pages 4 to 12
Lesson 2	AI Ethical Issues	Booklet 2, Unit 2	Pages 14 to 23
Lesson 3	AI and Future of Work (I)	Booklet 2, Unit 6	Pages 25 to 33
Lesson 4	Societal Impact of AI (I)	Booklet 2, Unit 7	Pages 35 to 45
Lesson 5	AI and Future of Work (II)	Booklet 3, Unit 4	Pages 47 to 58
Lesson 6	Societal Impact of AI (II)	Booklet 3, Unit 5	Pages 60 to 66

Note: Each lesson begins with a divider page, which is not part of the original booklet. Each original unit begins with a Lesson Plan and ends with an Assessment Exercise.

Lesson 1

AI Ethical Principles

Original source: Booklet 1, Unit 3

Original: 9 pages

Pages 4 to 12 in this booklet

The following pages are the original content, unchanged.

Unit 3 – AI Ethical Principles

Lesson Plan

Expected lesson time: 35 mins

After the lesson, students should be able to:

A. Knowledge and Skill:

1. Understand the AI ethical principles and apply them to various AI applications.

B. Attitude:

1. Realize that there are both advantages and disadvantages to AI applications and the need for ethical considerations for the development of AI.

Topic	*Task	Time (minutes in class / pre-lesson / after lesson)	Learning Resources
3.1 AI and Ethics	1	10 mins	
	2	12 mins	
	3	13 mins	Video

*Remarks

- Teachers can assign tasks flexibly (pre-lesson or in-class activities), and not all tasks are required to be covered during lesson time.
- Prior to playing any reference video in a unit, it is important for teachers to carefully review the content to ensure its suitability for the teaching purposes intended before proceeding with playing it.

Unit 3 – AI Ethical Principles

2.1 AI and Ethics

What is Ethics?

Ethics is a system of moral principles, considering:

- What is **good** versus **bad,** for individuals, groups and the society?
- What is **right** versus **wrong**?
- How can we balance our **rights** with **responsibilities?**
- How can our work benefit not only of **ourselves**, but also **others** and the **society?**

Ethical considerations of AI

- AI can help (i.e. do good) and AI can hurt (i.e. do harm)
- We must consider possible benefits versus harm, i.e. ethical considerations, in AI applications
- There are several globally converging principles among AI ethics guidelines

Task 1: Group Discussion: AI for Grading in Education

AI application can automatically and quickly grade students' assignments or examinations. AI grading models are trained on a large dataset of previously graded assignments, which include a diverse range of past answers and feedback from human graders. AI algorithms can now be used for grading examinations. What may be the advantage(s) or disadvantage(s) of AI grading in education?

Advantages:	
Disadvantages:	

This learning unit selects five most common ethical principles - transparency, justice and fairness, beneficence, responsibility, and privacy for discussion, see Figure 3.1.

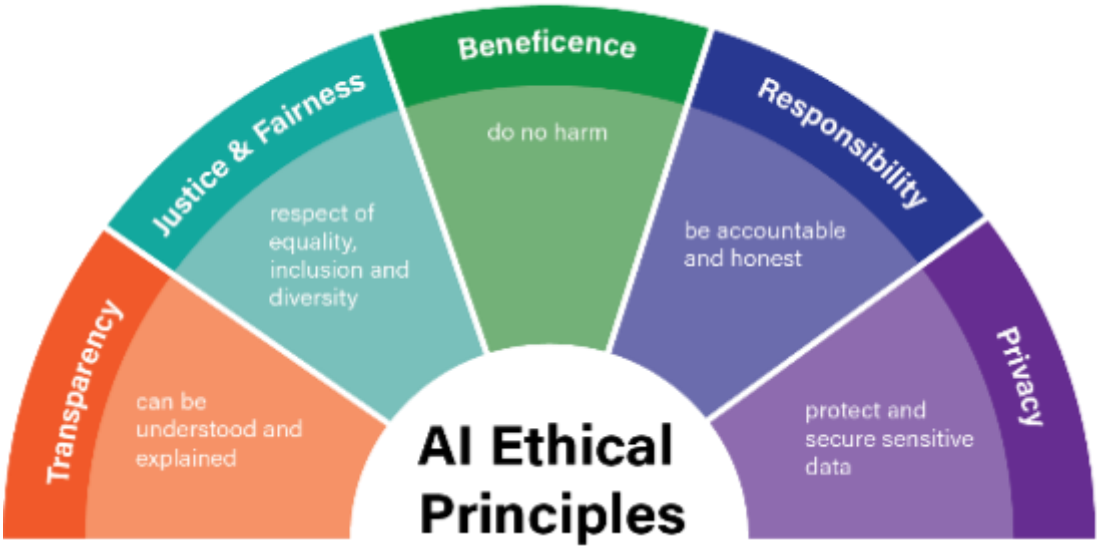


Figure 3.1: AI ethical principles

1. **Transparency:** Ensure that AI-enabled processing in applications can be interpretable by and explainable to users, enabling them to understand how decisions are made and the reasoning behind them.
2. **Justice and fairness:** Development of AI technologies and their applications must ensure equality, inclusion and diversity, by considering the impact of AI on different individuals and communities and avoiding biases and discrimination.
3. **Beneficence:** Development and application of AI technologies must not do any harm and should prioritise the safety and well-being of individuals, by minimising negative impacts on humans, society, and the environment.
4. **Responsibility:** Development and application of AI technologies must uphold accountability and honesty by considering the potential risks and taking responsibility for their impact on society.
5. **Privacy:** While developing AI technologies and incorporating them into real applications, sensitive data must be protected and secured by complying with relevant data protection and privacy regulations.

Task 2: Contact-tracing App during the COVID-19 Pandemic



Figure 3.2: Contact-tracing app

During the COVID-19 pandemic, people in Hong Kong use the contact-tracing app, LeaveHomeSafe, to record the date and time they visited different venues. This is done by scanning the venue's QR code to log the individual's arrival and clicking the "Leave" button in the APP to mark the departure. If a confirmed case of COVID-19 is later discovered at a venue, the APP will alert users who have visited the same venue at a similar time as the individual confirmed with COVID-19, as well as offer health advice.

The APP has recently been enriched with an AI-enabled functionality that can automatically record the time when users get out of a taxi, based on information captured by the accelerometer in the user's mobile phones. The AI technology performs this "auto-leave" task by monitoring the motion patterns of users. With the auto-leave feature, users do not have to tap the "Leave" button onscreen when they complete a taxi ride.

In the following table, please describe the ethical considerations in terms of transparency, justice & fairness, beneficence, responsibility and privacy from the aspect of the AI developers and the aspect of the general users.

Ethical Principles	Ethical Considerations for the AI Developer	Ethical Considerations for the General User
Transparency		
Justice and Fairness		
Beneficence		
Responsibility		
Privacy		

Task 3: What are the Possible Ethical Considerations that May Arise with the Use of AI Systems in the Domains of Autonomous Driving and Fake Video Generation?

Autonomous driving, also referred to as self-driving, is the operation of a vehicle without the need for direct human involvement. This advanced technology utilizes a range of advanced sensors, algorithms, and AI to sense and interpret the surrounding road and environment. As a result, autonomous driving systems can navigate and operate vehicle safely, efficiently, and without human intervention.



Figure 3.3: Autonomous driving (photo source from Adobe Stock)

Fake video generation involves the use of AI techniques, such as deep learning and computer vision, to create fabricated video content that appears to be genuine. Fake videos are used in a wide range of ways. Some are used for humorous or entertaining purposes, but others are used for political awareness, social media and other scam purposes.



Figure 3.4: Fake video generation (photo source from Adobe Stock)

In the table below, please describe the ethical considerations in terms of transparency, justice & fairness, beneficence, responsibility, and privacy for these two applications.

Ethical Principles	Autonomous Driving	Fake Video Generation
Transparency		
Justice and Fairness		
Beneficence		
Responsibility		
Privacy		

Learning Summary

AI ethics is a system of principles.

- These principles guide us to consider both the positive and negative impacts of AI when developing related technologies and using them in real applications.
- Ethical considerations are often very complex; thus, it is important to promote more thoughtful attention to the complex ethical problems.
- Controversies may arise about the future application of “Advanced AI”.

Unit 3 – Assessment Exercise

A. Fill in the blanks

1. "AI Ethics" is a system of principles that guide the consideration of both _____ and _____ impact in the development and usage of AI technologies.
2. Among the various AI ethical principles, _____ emphasises the importance of allowing people to know how AI technologies work.
3. Among the various AI ethical principles, _____ is important because sensitive data should be protected and secured.

B. Multiple-choice questions

1. Which ethical principle is violated if a company sells the personal data collected from users of their AI-powered applications without their consent?

   A. Transparency
   B. Responsibility
   C. Privacy
   D. Beneficence

2. Which of the following actions of an AI developer is NOT ethical?

   A. Providing details of the AI application to cover up the limitations of their products and services.
   B. Cross-checking a dataset for training AI models to ensure it is balanced.
   C. Using the data collected only for training and testing the AI application
   D. Ensuring their products and services will not cause any foreseeable or unintentional harm.

C. Short questions

1. List one example of how AI systems may raise ethical considerations.

_____

2. State the key concept of beneficence.

_____

3. If a smart self-driving car was in an accident with a person on the road, who is morally responsible?

_____

_____

_____

_____

Lesson 2

AI Ethical Issues

Original source: Booklet 2, Unit 2

Original: 10 pages

Pages 14 to 23 in this booklet

The following pages are the original content, unchanged.

Unit 2 – AI Ethical Issues

Lesson Plan

Expected lesson time: 35 mins

After the lesson, students should be able to:

A. Knowledge:

1. Understand the five common AI ethical principles (namely, Transparency, Justice and Fairness, Beneficence, Responsibility and Privacy).

B. Skills:

1. Apply the AI ethical principles and identify the possible risks in the use of AI technologies.

C. Attitudes:

1. Be aware of possible risks of AI technologies and applications.

Topic	*Task	Time (minutes in class / pre-lesson / after lesson)	Learning Resources
2.1 UNESCO Recommendation on the Ethics of AI		3 mins	
2.2 AI Principles and Ethical Issues			
2.2.1 Transparency		3 mins	
2.2.2 Justice and Fairness	1	8 mins	Video and website
2.2.3 Beneficence	2	5 mins	Video and website
2.2.4 Responsibility	3	10 mins	Video and website
2.2.5 Privacy	4	6 mins	

*Remarks

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- Prior to playing any reference video in a unit, it is important for teachers to carefully review the content to ensure its suitability for the teaching purposes intended before proceeding with playing it.

Unit 2 – AI Ethical Issues

2.1 UNESCO Recommendation on the Ethics of AI

AI is developing rapidly and shaping our future. The most important consideration in the development of AI is to ensure sustainable benefits – to be "human-friendly" and "earth-friendly". Recently, United Nation's Education, Scientific, and Cultural Organisation (UNESCO) has developed a guideline of ethical AI and provided the recommendations of AI ethical principles. The recommendations also outlined support of concrete policy actions to achieve the goal of developing ethical AI systems.

In the next section, we will have an in-depth discussion of issues related to five selected AI ethical principles – transparency, justice and fairness, beneficence, responsibility and privacy.

2.2 AI Principles and Ethical Issues

2.2.1 Transparency: AI transparency aims to ensure the explainability of the processes carried out by AI-powered applications for end users. It helps ensure that end users can understand how decisions are made, which promotes accountability and helps identify and address potential biases or errors in decision making.

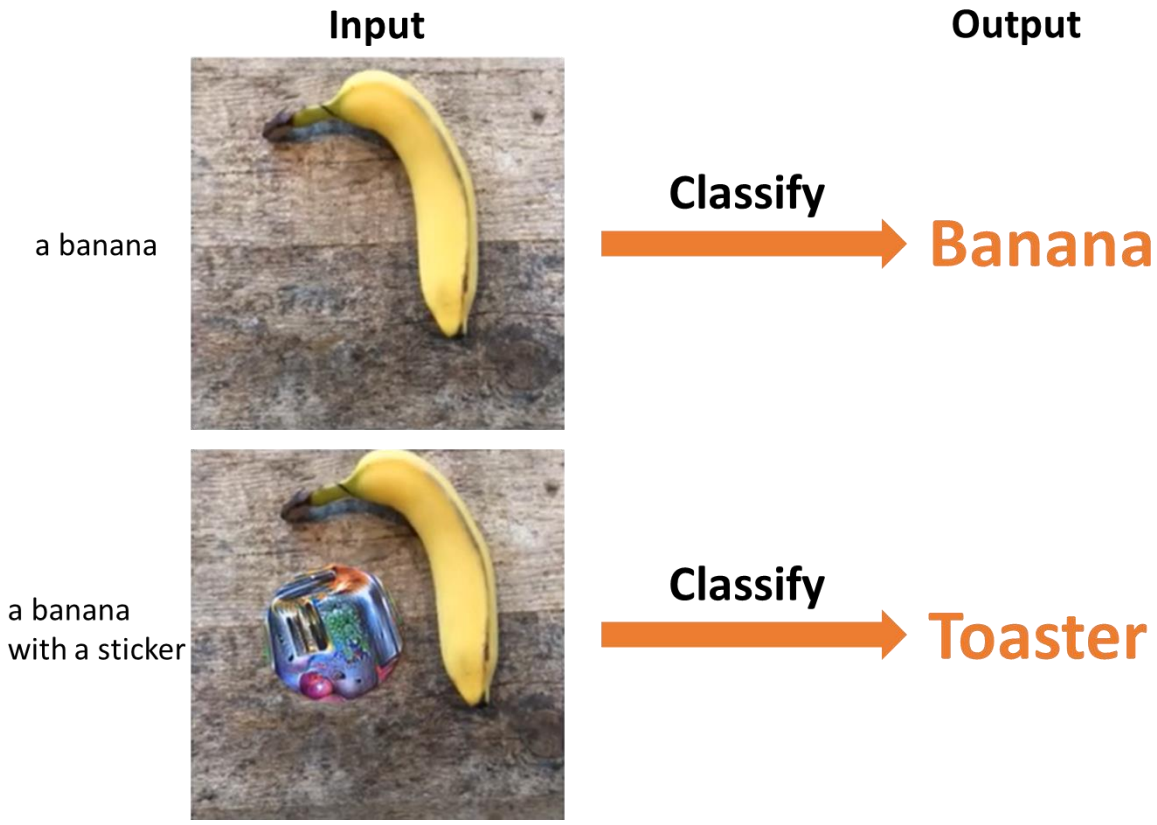


Figure 2.1: Image recognition of a banana, or a banana with a sticker.

AI models are often complex and lack transparency for tracing decision errors or correcting them. The above is an example from image recognition. As shown in Figure 2.1, if we show a banana alone to the image recognition model, it will recognize it as a banana. However, if we place a colourful sticker next to the banana, the model outputs the label "toaster". Presently, the image recognition model lacks transparency and it is difficult to explain why adding a sticker to the input image can cause this change. To counter these problems, the development of transparent AI models will be a key direction of the future.

2.2.2 Justice and Fairness: The development of AI technologies and their applications must ensure equality, inclusion and diversity. "Justice and fairness" can be considered as the rules of fair play. AI should promote justice and seek to eliminate all types of discrimination.

Task 1: AI for Screening Online Video Interviews

AI is often used to screen online interview videos for hiring staff. It analyses how you speak, the words you use and your emotional responses. Please read the materials and answer the following questions.

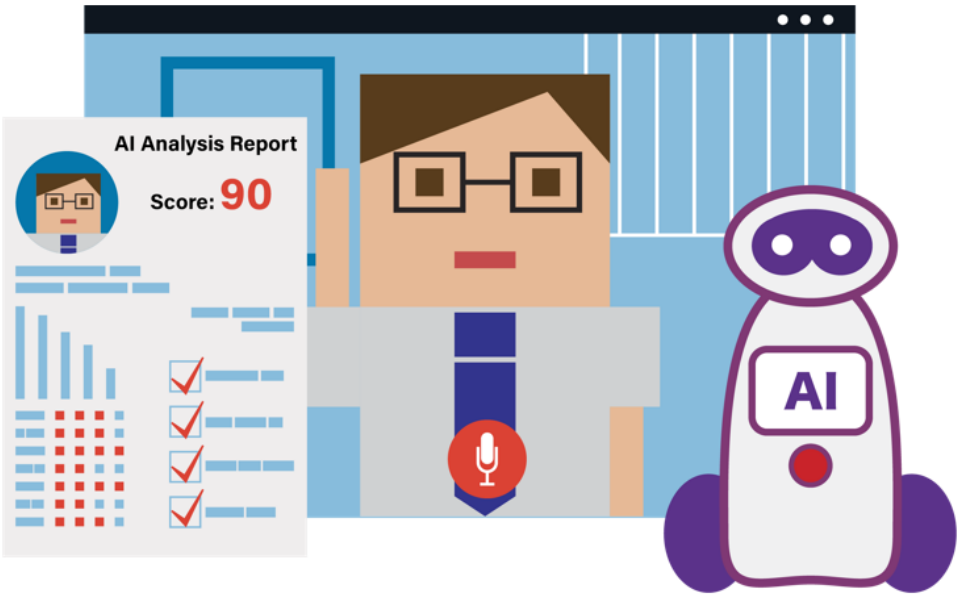


Figure 2.2: AI for screening online video interviews

https://www.recruiter.com/recruiting/pros-and-cons-of-using-ai-in-recruiting/

https://www.bbc.com/zhongwen/trad/world-55985943

What is/are the advantage(s) or disadvantage(s) of using AI for interviewing in hiring?

Disadvantage(s):

2.2.3 Beneficence: "Beneficence" refers to the development and application of AI technologies that must not do any harm. AI should promote well-being, preserve dignity, and enhance sustainability. AI should ultimately be developed for the common good and benefit of all humanity – "do good and do no harm".

Task 2: Airport Security Robots

AI can be used for security or military purposes. Please read the materials and answer the questions in the following section.



Figure 2.3: airport security robots

Airport Security Robots

https://www.ccb.or.jp/mice/e/news/securityrobots/

1. What are the functions of these robots?

2. What good does this do for society? Why?

2.2.4 Responsibility: "Responsibility" refers to the development and application of AI technologies that must uphold accountability and honesty. The development of AI systems and how they compare with the fundamental human values should ensure the well-being of humanity.

Task 3: Ethical Dilemmas Applied to Self-driving Cars

Real-world applications of AI may involve autonomous decision making, for example, how to avoid accidents in a self-driving car. Consider the classic trolley problem with a thought experiment that involves a moral dilemma - where an onlooker has the choice of saving 5 people in danger of being hit by a trolley, by diverting the trolley to kill just 1 person. What should be the trade-off in the onlooker's decision? Can there be sacrifices that are acceptable? If similar types of decisions need to be made by the self-driving car, what kind of moral choices should be programmed into the AI models? Please read the following materials and answer the questions below.

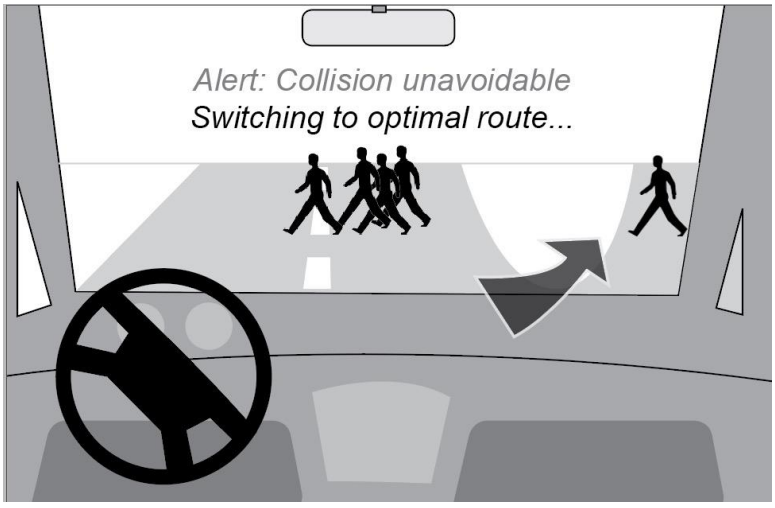


Figure 2.4: The ethical dilemma faced by a self-driving car (image source: harkeraquila.com, link)

Understanding Self-Driving Cars Under AI Technology – From Ethics to Privacy Issues

https://www.aili.com.tw/aili_detail/83.htm

The trolley problem

https://www.merriam-webster.com/words-at-play/trolley-problem-moral-philosophy-ethics

Moral Machine

https://www.moralmachine.net/

1. When programming a self-driving car to make decisions, what kind of ethical considerations need to be involved?

2. Who should be responsible for making the decisions?

2.2.5 Privacy: When AI technologies are being developed and incorporated into real applications, sensitive data must be protected and secured. AI systems create new privacy and ethical challenges. AI and machine learning consume massive volumes for training the AI model's decision-making process. However, there are numerous examples in which the power of AI algorithms results in privacy issues and ethical dilemmas. The principle of privacy should be to protect and secure sensitive data.

Task 4: Data Collected by Mobile Devices

Every time we use our mobile devices, we are releasing a lot of our personal data.



Figure 2.5: Mobile devices and data collection

1. What types of personal data can your phone collect about you?

2. When does your phone collect your data?

Learning Summary

- AI can help (i.e. do good) and AI can hurt (i.e. do harm). We need to consider possible benefits versus harm in AI applications.
- In November 2021, UNESCO's Recommendation on the Ethics of Artificial Intelligence was adopted by the 193 Member States as the first global standard.
- The ethical principles raise awareness of possible risks of AI technologies and applications and also guide the ethical development and development of AI to benefit humanity.

Unit 2 – Assessment Exercise

A. Fill in the blanks

1. Accidents involving self-driving cars raise concerns about accountability, which is related to the AI ethical principle of _____.
2. The use of artificial intelligence technology in surveillance and tracking may lead to problems with _____ .
3. In AI decision making, if algorithms do not treat certain groups of people equally, such as based on race, gender, or age, this can lead to a violation of the principles of _____.

B. Multiple-choice questions

1. Which of the following is an action in response to the principle of justice and fairness?

   A. Balance the data from female and male engineers to mitigate gender discrimination in AI hiring assistants.
   B. Make sure the applications do not bring any harm to people.
   C. Include illustrations and explanations of AI applications and products to help users better understand them.
   D. Protect users' personal information.

2. What is/are the possible outcome(s) for users in allowing virtual assistants to collect their personal information?

   (1) The virtual assistants will be able to provide more personalised recommendations for their users.
   (2) The AI technologies supporting the virtual assistant applications will be able to infer the users' daily routines based on their geolocation information.
   (3) Their personal information, if insufficiently protected, may be easily accessed by third parties without their knowledge.

   A. Only (1)
   B. (2) and (3)
   C. (1), (2) and (3)
   D. None of the above

C. Short question

1. What is the balance between the need for accuracy and transparency in the algorithms that people are seeking?

Lesson 3

AI and Future of Work (I)

Original source: Booklet 2, Unit 6

Original: 9 pages

Pages 25 to 33 in this booklet

The following pages are the original content, unchanged.

Unit 6 – AI and Future of Work (I)

Lesson Plan

Expected lesson time: 35 mins

After the lesson, students should be able to:

A. Knowledge:

1. Understand how technology changes the nature of work.
2. Learn about the characteristics of the four industrial revolutions.

B. Attitudes:

1. Be aware of the relationship between AI and work and develop a positive attitude to face the future of work.

Topic	*Task	Time (minutes in class / pre-lesson / after lesson)	Learning Resources
6.1 Past, Present and Future Jobs	1 & 2	10 mins	Article
6.2 Impact of Industrial Revolutions on Jobs	3 & 4	10 mins	Article
6.3 Future of Work with AI	5 & 6	15 mins	Article

*Remarks

- Teachers can assign tasks flexibly (pre-lesson, in-class, or assessment), and not all tasks are required to be covered during lesson time.
- Prior to playing any reference video in a unit, it is important for teachers to carefully review the content to ensure its suitability for the teaching purposes intended before proceeding with playing it.

Unit 6 – AI and Future of Work (I)

6.1 Past, Present and Future Jobs

AI is currently replacing humans in many tasks and such technological automation is replacing many jobs. Have you heard of these jobs (see Figure 6.1)?

AI, automation, and innovations are changing the nature of work as they re-invent business processes, operations and their execution. It is important to understand the future of work in different sectors.

Task 1: Identify Past, Present and Future Jobs

Which of the following jobs were common in the past? Which are common in the present? Which will be common in the future? Please put the labels in the table.



Figure 6.1: Different jobs

Past	
Present	
Future	

The nature of jobs keeps changing over the time. Technology is a driver for such changes. Some jobs no longer exist and have been replaced. We should not take the jobs we have now for granted or expect that they will last forever.

Task 2: Transformations in the Nature of Different Jobs

List the nature of the transitions in the following jobs, in the past, present and future.

Job	Past	Present	Future
E.g. Cashier	Humans record the list of sold goods by handwriting	Humans scan barcodes on goods	Humans occasionally help customers using self-checkout machines
Driver of a Vehicle			
Security Guard			

6.2 Impact of Industrial Revolutions on Jobs

Task 3: The Four Industrial Revolutions

Have you heard about the **four industrial revolutions**? They occurred when innovative technologies were developed and widely applied in industries:

Please scan the below QR codes for the references about the four industrial revolutions and complete the table:

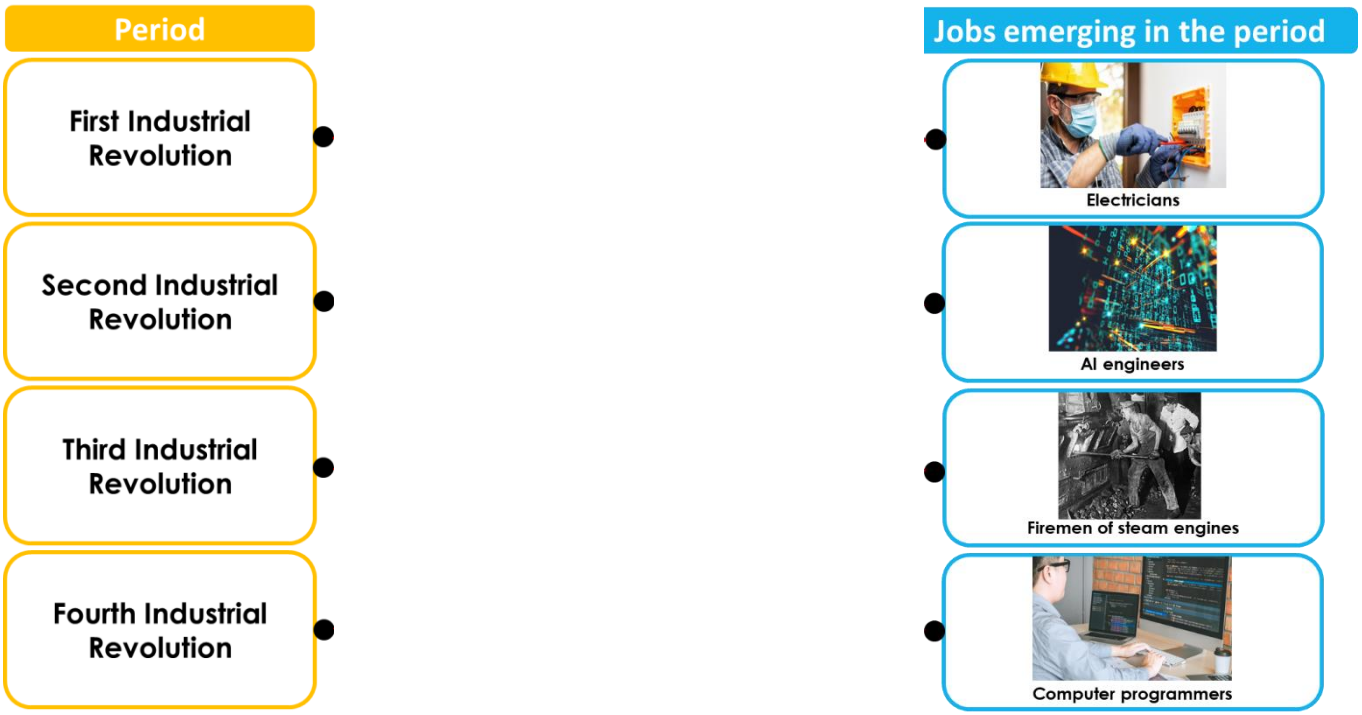
What is Industry 4.0? The fourth industrial revolution
https://sustainability-success.com/industry-1-0-to-4-0-2-3-revolution/#industry%204.0

(English version only)

Event	Characteristics
First Industrial Revolution (18th century)	• Technological breakthroughs: steam engines and metal forging • Manufacturing processes became mechanized
Second Industrial Revolution (19th century)	• Technological breakthroughs: electricity • Standardization, mass production, and industrialization • Mechanical production grew in speed
Third Industrial Revolution (20th century)	• Technological breakthroughs: computers, internet, and software • Software system carried out management processes
Fourth Industrial Revolution (Today)	

Task 4: Jobs and the Four Industrial Revolutions

Technology advancements in the four industrial revolutions have caused the disappearance of some jobs, and the creation of other jobs. According to the table in Task 3, can you try to match the period with the job emerging in the period?



6.3 Future of Work with AI

People have adapted to the changes brought by each industrial revolution. As jobs continue to be reshaped and changed by the emergence of new technologies, we should maintain a positive, inquisitive attitude and be prepared to learn new knowledge and acquire new skills. This will increase our competitiveness in the job market. Jobs related to developing and deploying new AI technologies are growing. New opportunities will increasingly emerge.

AI offers automation which can hopefully free up humans from jobs that are Dull (i.e. repetitive and tedious), Dirty, Dangerous and Dear (i.e. automation can do it faster and cheaper). Examples include cleaning, data entry, inspecting pollutants, etc.

Please scan the QR code to learn more about jobs AI may replace.

- The 7 jobs AI will most likely replace in 2022 – Automation is Here https://hudsonvalleystylemagazine.com/the-7-jobs-ai-will-most-likely-replace-in-2022/



- How many jobs have been replaced by technology? https://earthweb.com/how-many-jobs-have-been-replaced-by-technology/



- Will robots take my job? https://www.replacedbyrobot.info



Task 5: Jobs and Chatbots

A chatbot is an AI programme which aims to be able to engage in conversations with its users. Towards the end of 2022, a chatbot named ChatGPT was released by OpenAI and below we have listed some of its functions:

- Solve tricky math problems step-by-step
- Get relationship advice
- Write code
- Analyse market trends
- Write essays on almost any topic, e.g. children's book
- Summarise text
- Explain complex concepts
- Translation
- and more

While this chatbot can be a valuable tool for various industries, it is also important to consider both pros and cons of this technology.

What will happen if a chatbot like this one is applied in the workplace?

Task 6: Possible Advantages of Using AI in the Workplace

Should AI systems be used in the workplace? What are the potential advantages that AI may offer? Below we have listed some statements which try to describe the potential advantages of using AI in the workplace. Please indicate if you agree (✓) or disagree (✗), and give your reasons.

How AI may help in the workplace	Reasons	Agree/Disagree? (✓/✗)
Reduce workload for humans	AI systems can reduce the workload for humans. AI can replace tasks which are labour-intensive, repetitive and with low skill requirements. Humans will have more time to work on more interesting and even challenging tasks.	
Understand what is important to humans	AI is a technology and unlike humans, it does not have a sense of right versus wrong, good versus bad, ethical versus unethical.	
Make faster decisions	AI systems can make faster decisions with well-trained models and powerful computing resources.	
Benefit people	AI systems can potentially benefit people. AI can facilitate decision-making, enhance efficiency, reduce human errors and assist humans in our daily lives. For example, AI can assist with cancer detection from medical images, and AI security systems can be used in 24-hours CCTV.	
Improving performance	The increasing use of AI will help generate data for training better AI models. This may, in turn, encourage even greater use of AI and incentivize more people to learn about AI technologies.	
AI can do everything	Most AI systems have specific capabilities, because AI models are trained to focus on a targeted task. For example, AlphaGo knows how to play the game of Go, but not other board games. An AI system trained to diagnose one type of cancer for medical images may not be able to diagnose other types of cancers.	

Learning Summary

- AI is reshaping the future of work across different industries.
- The fourth industrial revolution is defined as a combination of advances in AI, robotics, the internet of things (IoT), and other new technologies.
- Keep learning and stay positive to face the AI future of work.

Unit 6 – Assessment Exercise

A. Fill in the blanks

1. The combination of advances in AI, robotics, the internet of things (IoT), genetic engineering, quantum computing, and other new technologies have brought about the _____ industrial revolution.
2. We should maintain a positive, inquisitive attitude and be prepared to learn new knowledge and acquire new skills. This will increase our _____ in the job market.

B. Multiple-choice questions

1. Which of the following statements is true?

   A. All jobs will continue to exist as they are now and will not change.
   B. AI will replace all human jobs.
   C. When introducing AI technologies in the workplace, productivity is more important than ensuring ethical use.
   D. AI may replace some jobs and assist human in other jobs.

2. Please match the job with the corresponding period:

Job		Period
1.	AI engineers	i. First industrial revolution
2.	Computer programmers	ii. Second industrial revolution
3.	Electricians	iii. Third industrial revolution
4.	Firemen of steam engines	iv. Fourth industrial revolution

   A. 1: i; 2: ii; 3: iii; 4: iv
   B. 1: iv; 2: iii; 3: i; 4: ii
   C. 1: iv; 2: iii; 3: ii; 4: i
   D. 1: iii; 2: iv; 3: i; 4: ii

C. Short questions

1. List one job that is MOST likely be replaced by AI in the near future.

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2. List one job that is LEAST likely be replaced by AI in the near future.

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3. How can individuals prepare for the future of work let by AI?

Lesson 4

Societal Impact of AI (I)

Original source: Booklet 2, Unit 7

Original: 11 pages

Pages 35 to 45 in this booklet

The following pages are the original content, unchanged.

Unit 7 – Societal Impact of AI (I)

Lesson Plan

Expected lesson time: 35 mins

After the lesson, students should be able to:

A. Knowledge:

1. Learn about the benefits and harms of AI in our daily lives.

B. Skills:

1. Identify examples of social good brought about by AI.
2. Understand and apply the Five Steps of Design Thinking.

Topic	*Task	Time (minutes in class / pre-lesson / after lesson)	Learning Resources
7.1 What is Social Good?	1 & 2	10 mins	Video
7.2 Examples of AI for Social Good	3	7 mins	
7.3 Examples of AI becoming a Detriment to Society		8 mins	Online Application
7.4 How to Design AI for Social Good	4	10 mins	Article

*Remarks

- Teachers can assign tasks flexibly (pre-lesson, in-class, or assessment), and not all tasks are required to be covered during lesson time.
- Prior to playing any reference video in a unit, it is important for teachers to carefully review the content to ensure its suitability for the teaching purposes intended before proceeding with playing it.

Unit 7 – Societal Impact of AI (I)

7.1 What is Social Good?

Social good refers to something that benefit the public. Ideally, technologies should promote social good, i.e. provide benefits and spread them widely. However, this depends on how properly the technologies are designed and managed. AI can perform many tasks better than humans, which can be either beneficial or harmful.

Task 1: AI for Social Good

Is AI beneficial for society? ..... Yes / No

Could you give some examples of benefits or harms in using AI?

Potential benefits:
Potential harms:

Task 2: The TrailGuard Artificial Intelligence Camera

Please scan the QR code to watch the video and discuss the following questions in a group:
https://gocuhk-my.sharepoint.com/:v:/g/personal/b194474_cuhk_edu_hk/Efpq-qCMRGpHhH2W6XJZnJUBCLV6yuh6VXVIzHpdUrz-6g?e=vn1CM2

(Video Sources：Intel Corporation)

1. What is the problem that needs to be solved? How is it solved?

2. What AI technologies are being used?

3. Is it a good design? Why?

7.2 Examples of AI for Social Good

Task 3: More Examples of AI for Social Good

Below are three examples of AI for social good. How do they bring benefits to society?

Examples	Benefits to the society
1. AI in COVID-19	
2. Chatbot "Bonny"	
3. Digital mental health screening	

AI in COVID-19

In 2021, a research team from The Chinese University of Hong Kong developed an AI system to detect COVID-19 automatically, quickly and accurately based on 3D chest CT images. During COVID-19, the workload to analyse and interpret medical image data has been unprecedentedly large and urgent. Such an AI system can effectively facilitate the management of COVID-19.

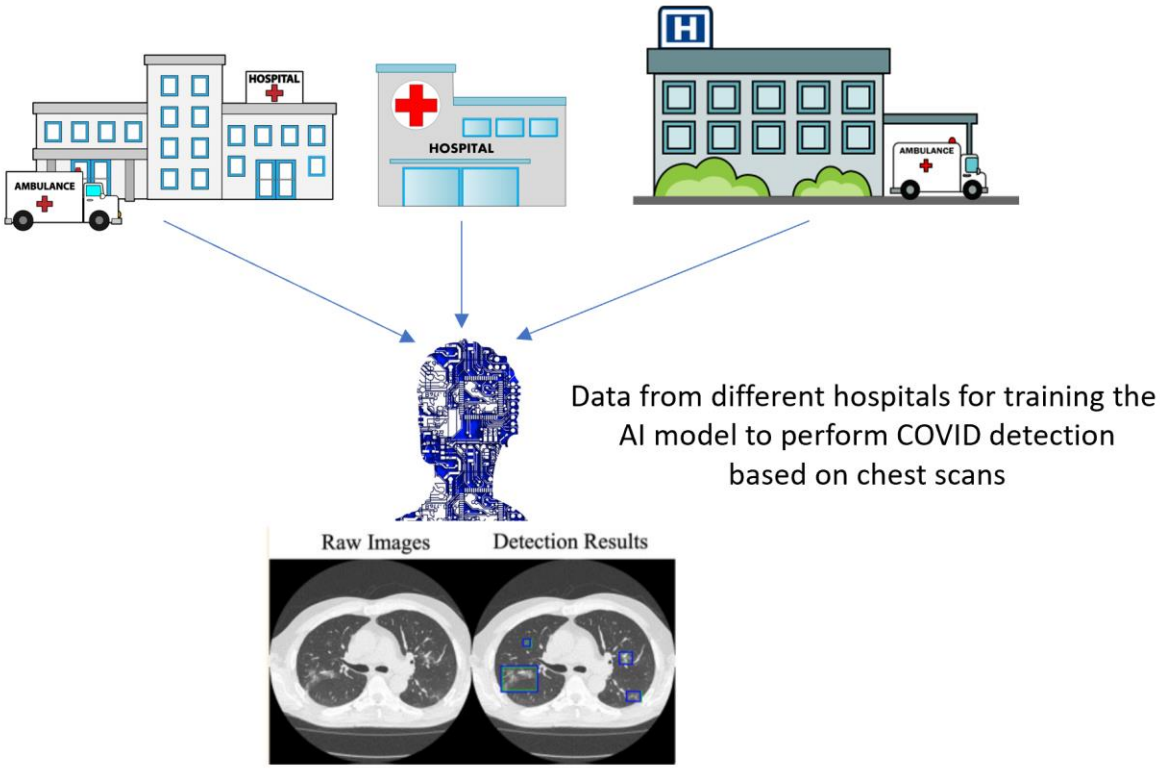


Figure 7.1: AI facilitates COVID-19 medical image analysis

Chatbot "Bonny"

"Bonny" is a chatbot launched in December 2019 on the website of the HKSAR Government. It helps users conduct searches for government e-services, application forms and related URL links. "Bonny" is able to respond in both Chinese and English and provide relevant information through an online chat box.

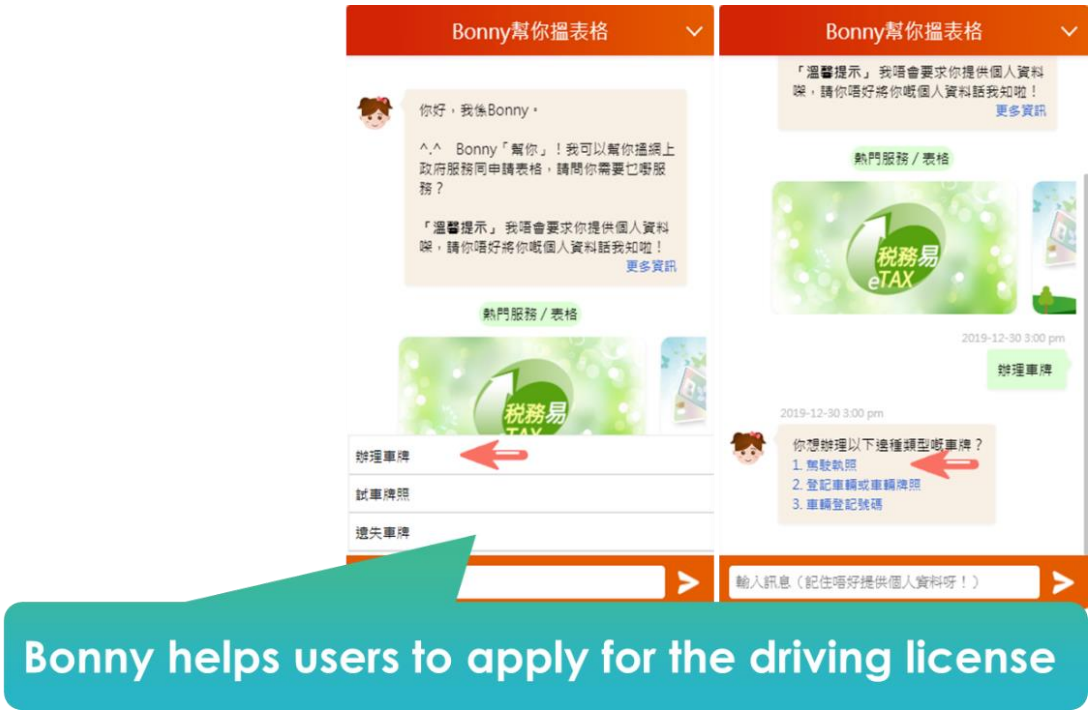


Figure 7.2: Chatbot “Bonny”

Digital Mental Health Screening for Secondary Schools

“Know My Students” is a digital mental health screening program developed by a local clinical psychologist. It is a digital conversation tool that applies AI technologies to help schools identify and support students who are struggling with their mental health.



Figure 7.3: Digital health screening for secondary school students (source: https://www.knowmystudents.com/)

7.3 Examples of AI becoming a Detriment to Society

In previous units, we have experienced the power of ChatGPT.

AI may Lead to Complacency

Do you think AI technologies like ChatGPT will cause students to become complacent and dependent completely on ChatGPT to generate answers for assignments? Will complacency lead us to give up our ability and control in finding complex solutions to difficult problems? Please discuss.

Please scan the QR code to access ChatGPT:
https://poe.com/login?redirect_url=%2F

AI may Threaten Jobs

Also, do you think advances in AI technologies will lead to the loss of many jobs? In which jobs do you think AI will replace humans? (Please refer to Booklet 2 Unit 6 and Booklet 3 Unit 4 Future of Work for more discussions on how AI may threaten our jobs)



Figure 7.4: Will AI replace humans in graphics design? (Image generated by Midjourney)

AI may Collect Sensitive Information

Users must be very careful when using AI, because any data you provide or upload may be kept and used for other purposes. Please read the news about a company's workers using AI to help them with tasks and unintentionally leaked the company's top secrets.

News: Samsung workers made a major error by using ChatGPT
https://www.techradar.com/news/samsung-workers-leaked-company-secrets-by-using-chatgpt

7.4 How to Design AI for Social Good

AI is often regarded as a key contributor to the fourth industrial revolution. The disruption from an industrial revolution tends to cause many people to lose their jobs. Hence, we must prepare ourselves well for the future as we live in the age of AI. To master AI technologies, it is important to have foundations in mathematics and science and develop strong critical and creative thinking skills, especially in design thinking.

Design thinking is the creative use of existing knowledge or technologies to address emerging problems. Design thinking creates various ways of using AI technologies, designing for the benefit of others will help us develop AI for the right reasons and become a better person.

The Five Steps of Design Thinking

We are going to learn the five steps of design thinking. The steps of design thinking can be divided into five key steps: Empathize, Define, Ideate, Prototype and Test, see Figure 7.5.

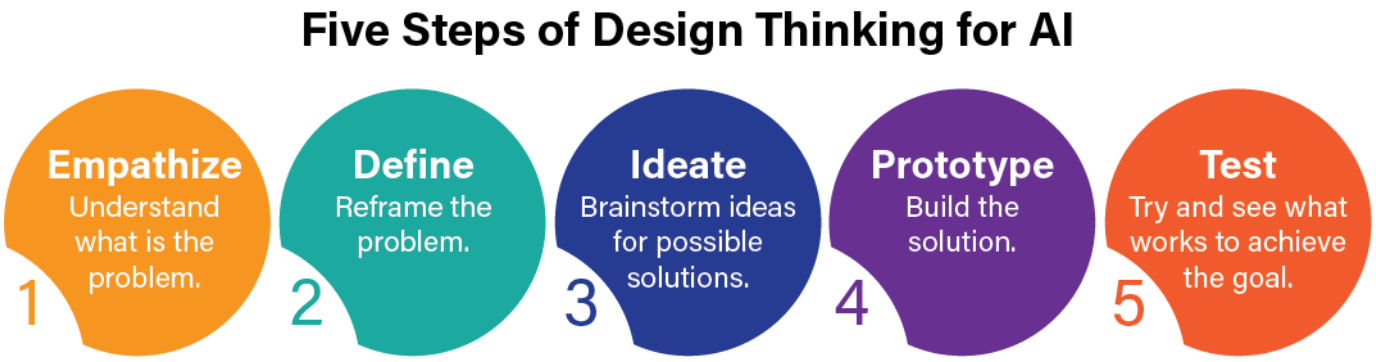


Figure 7.5: The Five Steps of Design Thinking of AI

Please scan the QR code for more details about the five steps of design thinking:
5 Steps of the Design Thinking Process: A Step-by-Step Guide
https://voltagecontrol.com/blog/5-steps-of-the-design-thinking-process-a-step-by-step-guide/

Task 4: Improve Swimming Pool Safety by Adopting the Five Steps of Design Thinking

Now that we have learnt about the Five Steps of Design Thinking, we can apply it in a real-life situation – swimming pool safety. Using the design thinking approach, we will generate a real problem statement and a feasible solution in order to achieve social good with AI.

Step 1: Empathize

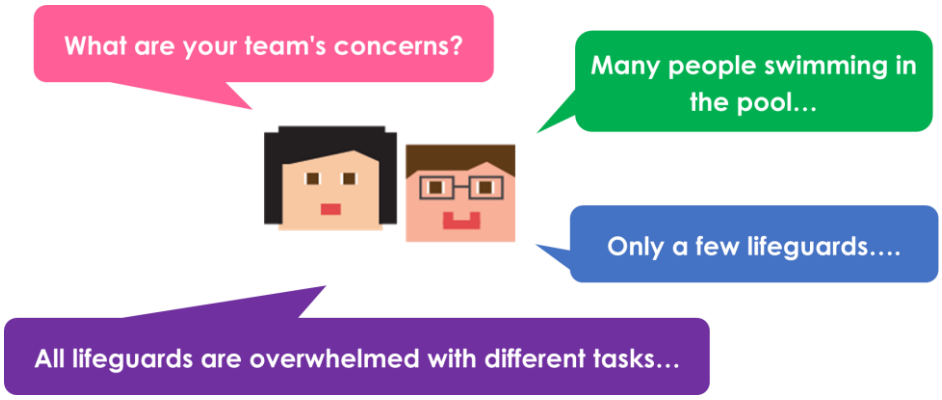


Figure 7.6: Empathize step

What are your team's concerns?

Step 2: Define

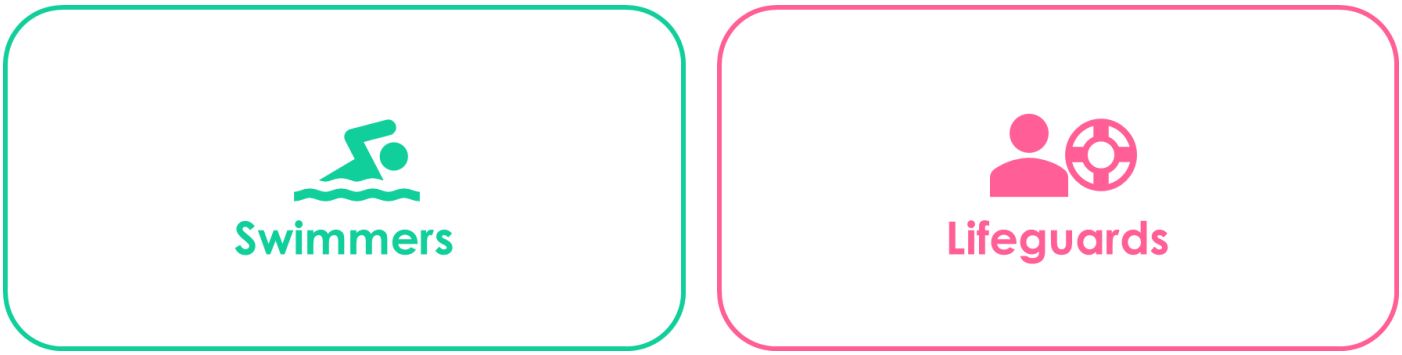


Figure 7.7: Define step

Who are affected and how are they affected?

Swimmers:

Lifeguards:

Step 3: Ideate

How do you use AI to solve problems?

- When should AI technologies be used to solve defined problems?
- Suggest how AI technologies can solve the problems.
- Imagine how the application of AI technologies can solve problems.

(Hints: What kind of functions should AI have for solving the problems?)

Step 4: Prototype

When you design an AI lifeguard system,

- What sensory inputs does the system require?
- To train the AI system, what data do you need and where can you obtain the data? What are the challenges here?
- How do you implement your design?

Step 5: Test

Propose how you would test your ideas. *(Hints: Would it be better to install the system under water? Could it be used for coaching as well?)*

Learning Summary

- AI may bring benefits or cause harm to humans, so we must use the technologies carefully.
- AI technologies can provide new ways to promote social good and improve the quality of peoples' lives.
- Design thinking is a possible approach to devise AI-enabled solutions in an attempt to solve (complex) problems.
- The five steps of design thinking: empathize, define, ideate, prototype and test.

Unit 7 – Assessment Exercise

A. Fill in the blanks

1. Social good refers to things or events that can _____ the public widely.
2. In the Five Steps of Design Thinking, the first step is to _____ , which aims to understand the problem.
3. In the Five Steps of Design Thinking, the fourth step is to_____ , which aims to make a sample solution that users can experiment on.
4. A good AI design should devote considerations to ethical use and possible _____ impact.

B. Multiple-choice questions

1. What is the last step in the Five Steps of Design Thinking?

   A. Empathize
   B. Define
   C. Prototype
   D. Test

2. What can AI technologies do for us in the future?

   A. AI can help improve our lives and support us in our work
   B. AI may present risks and even dangers to humans
   C. Both A and B
   D. None of the above

C. Short questions

1. List three examples of AI for social good.

_____

2. List the Five Steps of Design Thinking.

_____

Lesson 5

AI and Future of Work (II)

Original source: Booklet 3, Unit 4

Original: 12 pages

Pages 47 to 58 in this booklet

The following pages are the original content, unchanged.

Unit 4 – AI and Future of Work (II)

Lesson Plan

Expected lesson time: 35 mins

After the lesson, students should be able to:

A. Knowledge:

1. Understand how AI changes the future of work.

B. Skills:

1. Outline the advantages and disadvantages brought by AI.

C. Attitudes:

1. Be well-prepared to develop skills that meet the need of the future job market.

Topic	*Task	Time (minutes in class / pre-lesson / after lesson)	Learning Resources
4.1 AI Changing the Future of Work	1	7 mins	Online resources
4.2 Experiencing Generative AI Technologies	2 & 3	10 mins	Online resources
4.3 How AI will Change Jobs in Industries		6 mins	Online resources
4.4 Jobs with Increasing / Decreasing Demands	4	6 mins	
4.5 Skills to be Developed for the Future of Work		6 mins	Online resources

*Remarks

- Teachers can assign tasks flexibly (pre-lesson, in-class, or assessment), and not all tasks are required to be covered during lesson time.
- Prior to playing any reference video in a unit, it is important for teachers to carefully review the content to ensure its suitability for the teaching purposes intended before proceeding with playing it.

Unit 4 – AI and Future of Work (II)

4.1 AI Changing the Future of Work

The rapid advancement of AI is changing the future of work. AI technologies can automate many tasks, but the technologies are not perfect. Therefore, while some jobs have great automation risks, the technologies cannot entirely replace humans in other jobs. Instead, they are changing the nature of the jobs and creating new opportunities.

Will AI Take Your Future Job?

Which industry as well as type of work would you like to pursue in the future? You can visit this website and select a few examples to discover which jobs may be affected by AI.

Link for the website:
https://willrobotstakemyjob.com

Task 1: Risks of AI-Automation in Different Jobs

In the following, we list out different jobs. Arrange them in order of decreasing risk of AI-automation. Discuss with your classmates and explain your answers.



Figure 4.1: A Variety of Jobs

4.2 Experiencing Generative AI Technologies

The recent, rapid developments of Generative AI technologies brought powerful applications such as ChatGPT, released by OpenAI in November 2022. A more advanced model, GPT4, was released in March 2023. These models are trained on a vast amount of data on the internet, including hundreds of billions of words from Wikipedia, books, news articles, journals, and other sources. Hence the models can generate language and code. The application attempts to conduct a human-like dialog with the user. You may access ChatGPT and try it in Task 2:

Task 2: Experiencing Generative AI through ChatGPT

You may access a website related to ChatGPT (e.g. www.poe.com), input a request, and see what it can do for you. For example, you may consider seeking help from the AI technology to write a cover letter for a job application (see left part of Figure 4.2):

Upon receiving the input request, we found that ChatGPT can immediately generate the output below (see right part of Figure 4.2):

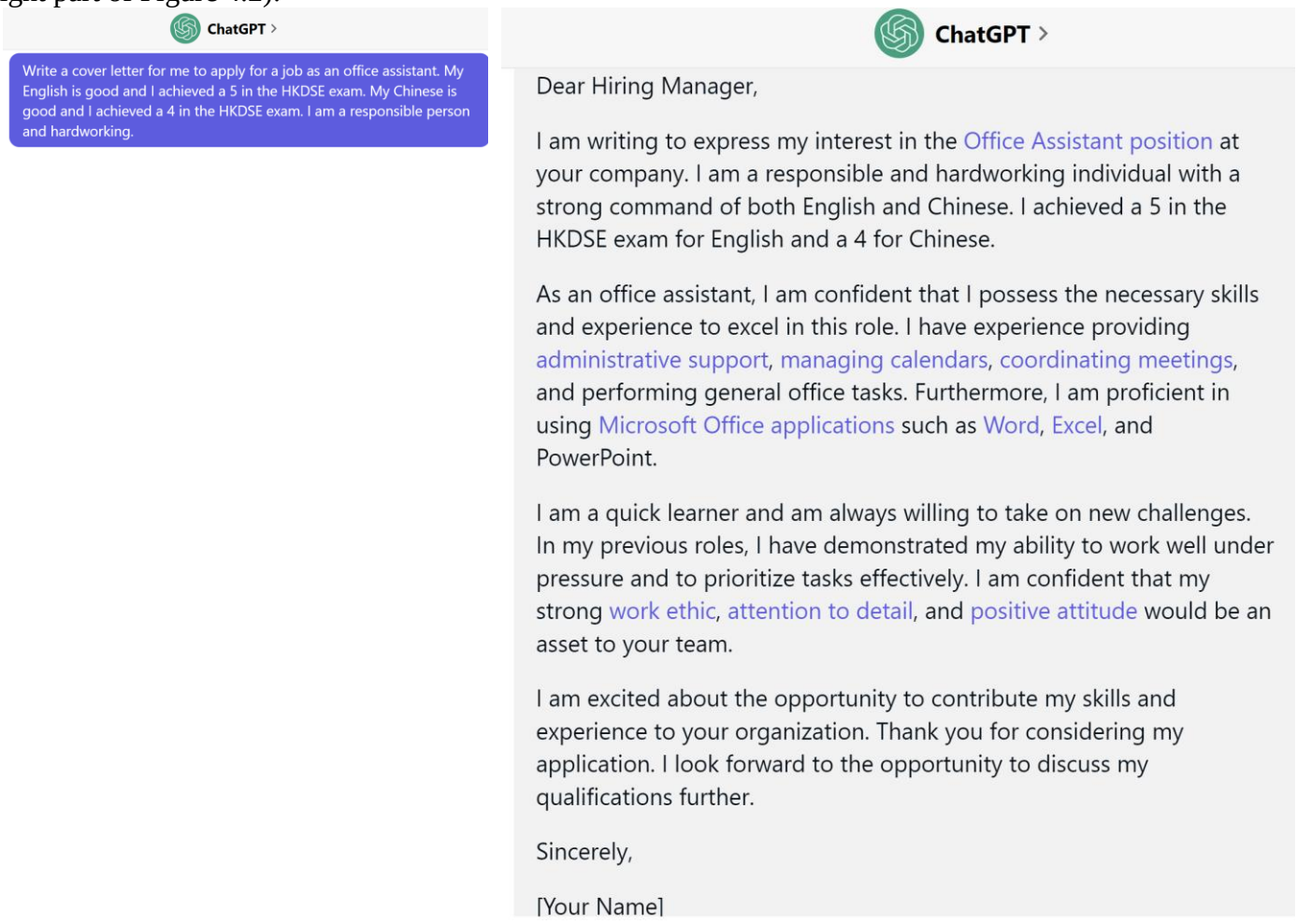


Figure 4.2 Response of ChatGPT to the request for writing a letter.

What is your opinion of the quality of the letter? Is all the information in the letter correct? Try to input another request into the AI system and discuss the quality of the output. If such a kind of technology is used in the office, how will it change the job of an office worker?

What are the other requests that you input into the system? What do you think of the outputs? Do they satisfy your needs? What are the advantages and disadvantages of this AI technology?

Generative AI technologies can also generate images from text. Some applications include Dall-E, Midjourney, DreamStudio (Stable Diffusion), etc. Users may input a text prompt and the system will generate related images immediately. For example, we may try to automatically generate a company logo using the AI technology, as will be illustrated in Task 3.

Task 3: Experiencing Generative AI through DreamStudio

You may access a related website (e.g., https://beta.dreamstudio.ai/generate), input a request, and see what it can do for you. For example, you may consider generating a new company logo for an AI company, using green as the theme colour, and bring out the innovative characteristics of the company, which is located in Hong Kong (see left part of Figure 4.3). Almost immediately, the application returned with a few generated images (see right part of Figure 4.3).

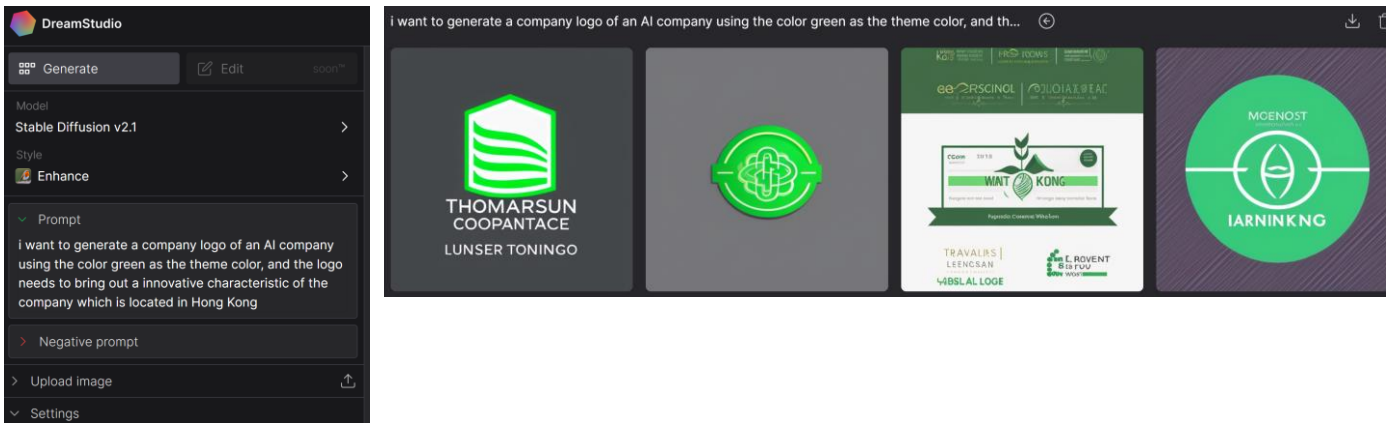


Figure 4.3 Response of DreamStudio to the request for drawing a company logo.

What is your opinion of the quality of the generated logos? Is all the information in the logos correct? Try to input another request into the AI system and discuss the quality of the output. If you were a logo designer, how will you make use of this kind of Generative AI technology?

What are the other requests that you input into the system? What do you think of the outputs? Do they satisfy your needs? What are the advantages and disadvantages of this AI technology?

4.3 How AI will Change Jobs in Industries

In the following, we will discuss how AI may change the jobs in a few example industries.

Banking and Finance Industry

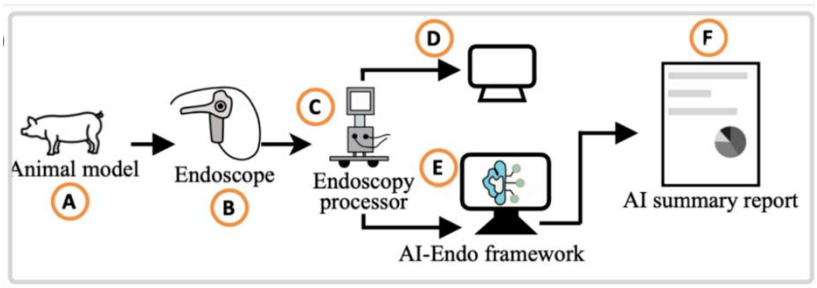
Many banks and financial institutions are increasing their adoption of AI technologies. For example, AI-enabled chatbot technologies to serve their customers in a more personalized way while minimizing their waiting time. The conversational nature of the technologies are generally favoured by customers. The recommendation systems enabled by AI can introduce more personalized products and services for the customer. Also, the AI can capture each individual's spending behaviour and help prevent fraud, e.g. credit card fraud detection based on anomaly detection.

However, fully automated services AI may not be sufficiently robust and error-free, especially for important tasks such as protecting the customer's personal financial data, providing the human touch in services that is preferred by many customers. It may even be possible that the AI technologies make biased decisions, e.g. discriminate against customers with lower income as they offer service support. Therefore, workers of the future in the industry of banking and finance must be skilled in using AI technologies in their work and knowledgeable about the strengths and weaknesses of the AI technologies.

[Think about it] Can you give an example of the use of AI in the banking and finance industry? What advantages and disadvantages will AI bring? What are the skills that we should develop for the jobs of the future in banking and finance?

Healthcare Industry

The healthcare industry is also increasingly adopting AI technologies. For example, AI can speed up drug discovery for diseases by using image analyses that focus on molecules to determine drug effects [Johnson&Johnson]. Also AI can be used in surgery (see Figure 4.4) to analyse complex anatomical scenes from vast amounts of surgical data, to model the stages in the surgical process for investigating procedural knowledge and decision-making skills, reduce complications and enhance surgical safety.



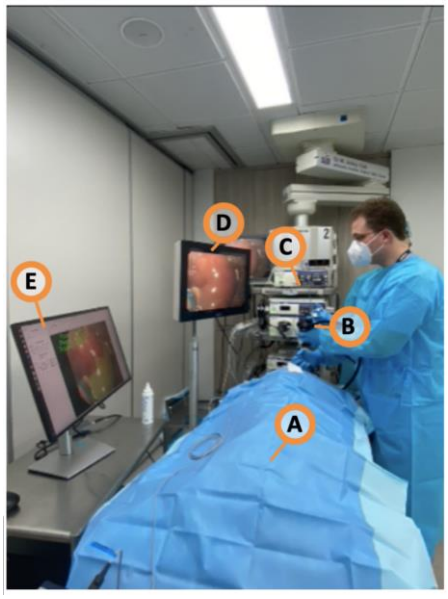


Figure 4.4: Experimental settings (left) and real-time performance (right) of pre-clinical animal experiments in the use of AI during endoscopic surgery using the clinical Olympus Endoscopic system.

(A) is the animal model. (B) is the endoscope. (C) is the endoscopy processor for video streaming. (D) is the Olympus screen. (E) is the AI system's screen displaying analytical results. (F) is the AI-generated summary report on the results of surgical phase recognition. [Source: AI-Endo, CUHK]

We must be mindful that AI technologies have imperfections. We must be extremely careful about incorporating AI in the healthcare industry, because AI biases or errors may lead to life-threatening situations. For example, we must ensure that there is an experienced medical expert overseeing the automated processes. Therefore, workers of the future in the healthcare industry must acquire the skills of using AI technologies in their work and be knowledgeable about the strengths and weaknesses of the AI technologies. With such knowledge and skills, they may be able to utilize AI as a tool for decision support for medical professionals in healthcare applications, and the partial automation will increase their capacity in delivering better care for more patients.

[Think about it] Can you give an example of the use of AI in the healthcare industry? What advantages and disadvantages will AI bring? What are the skills that we should develop for the jobs of the future in the healthcare industry?

[AI-Endo reference] paper will be published soon.

Retail and Marketing Industry

AI has many possible applications in retail and marketing. Since the COVID-19 pandemic started in 2020, we are seeing that an increasing number of shopping malls use robots for cleaning and sanitisation. These robots can cover the full shopping area and avoid customers who are walking around. There are also shops that can track every customer after he or she logs in and enters a store, and records the items that are selected, as well as charges to the customer's account upon departure. This way, the customer does not need to wait in line for payment [Amazon Go]. Computer vision can also be used to monitor the store's shelves and order replenishment when a product is running low. Generative AI can also be used by marketers to compose marketing prose, as illustrated in the example in Figure 4.5.

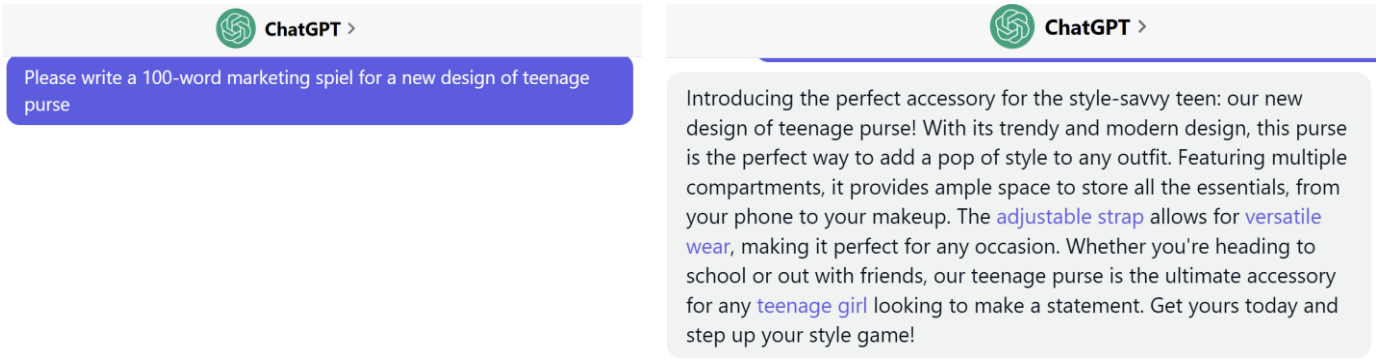


Figure 4.5 The use of Generative AI to compose marketing text.

AI will continue to transform the retail and marketing industry, bringing increased productivity and efficiency, as well as convenience to customers. It is important to remember that AI has limitations and may be affected by data bias, model bias or errors. It will be desirable to ensure that the business processes are overseen by humans. Also, when Generative AI tools are used for generating content in marketing, it will also be important for the human to incorporate his/her own creativity into the writing, which is necessary to bring out the unique characteristics of the products and services.

[Think about it] Can you give an example of the use of AI in the retail and marketing industry? What advantages and disadvantages will AI bring? What are the skills that we should develop for the jobs of the future in the retail and marketing industry?

[Think about it] Can you give an example of the use of AI in industries other than the above? What advantages and disadvantages will AI bring? What are the skills that we should develop for the jobs of the future?

4.4 Jobs with Increasing / Decreasing Demands

According to a report from the World Economic Forum's "The Future of Jobs Report 2020", AI automation will replace millions of existing jobs and create millions of new jobs by 2025.

Task 4: Jobs with Increasing Demand and Jobs with Deceasing Demand in 2025

Can you guess in the year of 2025, compared to now, the demand of which of the following jobs will be increasing? Which will be decreasing?

- Information Security Analysts
- Client Information and Customer Service Workers
- Material-Recording and Stock-Keeping Clerks
- Big Data Specialists
- General and Operations Managers
- Data Analysts and Scientists
- Data Entry Clerks
- AI and Machine Learning Specialists
- Internet of Things Specialists
- Assembly and Factory Workers
- Software and Applications Developers

Jobs with increasing demand	Jobs with decreasing demand

4.5 Skills to be Developed for the Future of Work

Although certain tasks can be replaced by AI, we should nevertheless be well prepared and further develop skills that cannot be replaced by AI. To be competitive in the future job market, OECD (The Organization for Economic Cooperation and Development) project "Future of Education and Skills 2030" proposed three classes of skills that students should develop:

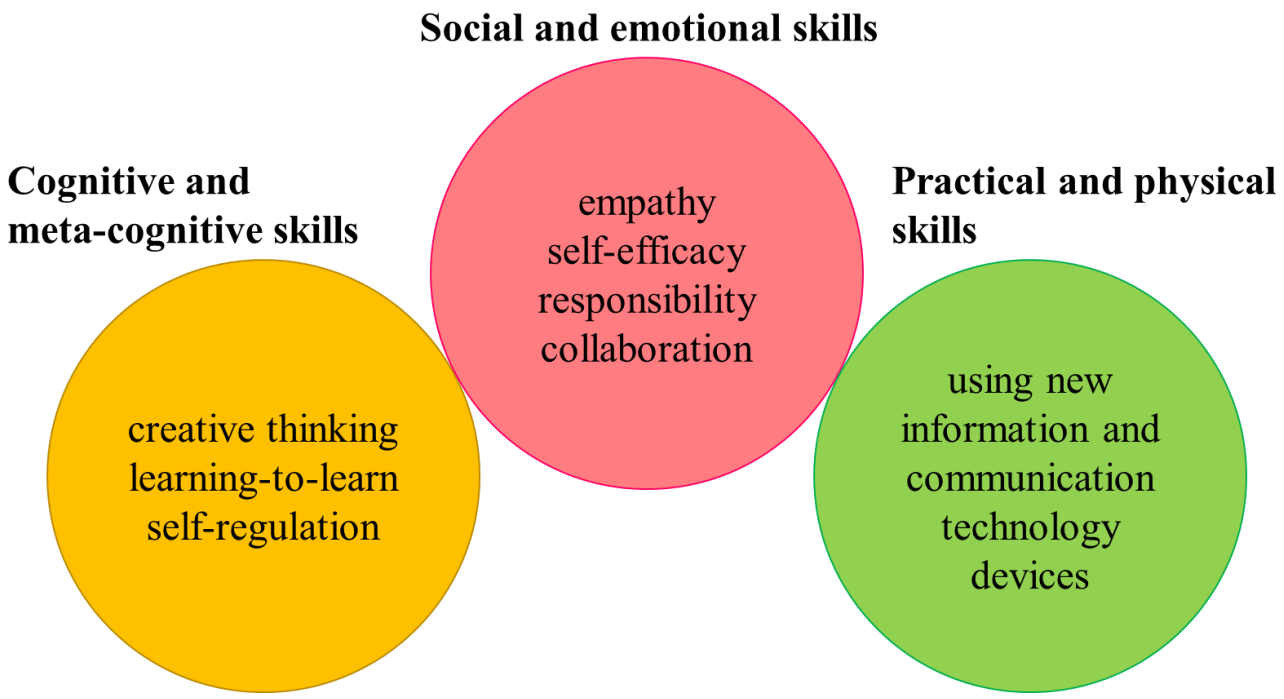


Figure 4.6: Skills to be developed for the future of work

How would you prepare yourself in order to meet the need of the future job market?

_____

_____

Learning Summary

- AI is changing the future of work. AI brings increasing automation, together with advantages and disadvantages.
- We should develop skills to utilize AI in different jobs.
- We must also develop skills that cannot be replaced by AI, including social and emotional skills, cognitive and meta-cognitive skills, as well as practical and physical skills.

Unit 4 – Assessment Exercise

A. Fill in the blanks

1. AI brings increasing _____ to different tasks, which changes the jobs of the future.
2. We need to be careful about total automation of tasks using AI, because the technologies may not perform perfectly due to _____, _____ and _____ .
3. Recently, _____ AI technology such as ChatGPT has become increasingly popular due to its ability to generate human-like language and perform various tasks.

B. Multiple-choice questions

1. How is the rapid advancement of AI changing the future of work?

   A. AI is replacing all human jobs with automation
   B. AI is creating entirely new jobs that humans have never done before
   C. AI is changing the nature of jobs and creating new opportunities, while some jobs have great automation risks
   D. AI is not changing the future of work

2. What is the purpose of developing skills that cannot be replaced by AI in different jobs?

   A. To eliminate the need for human labor
   B. To ensure that human can work with AI in different jobs
   C. To create a dependence on AI
   D. To reduce the job opportunities

3. Which skills will become very important for future jobs with AI-automation?

   A. Social and emotional skills
   B. Cognitive and meta-cognitive skills
   C. Practical and physical skills
   D. Programming skills

C. Short questions

1. What is your dream job? Can AI be applied in your dream job? How?

2. What are some ethical issues that need to be concerned when applying AI in your dream job?

Lesson 6

Societal Impact of AI (II)

Original source: Booklet 3, Unit 5

Original: 7 pages

Pages 60 to 66 in this booklet

The following pages are the original content, unchanged.

Unit 5 – Societal Impact of AI (II)

Lesson Plan

Expected lesson time: 35 mins

After the lesson, students should be able to:

A. Knowledge:

1. Understand how using AI for social good can contribute to the Sustainable Development Goals (SDGs) and solve challenges that have an impact on our society, economy, and environment.
2. Learn about generative AI models such as ChatGPT.

B. Skills:

1. To analyze how AI will affect society and understand the importance of making sure the use of AI is aligned with ethical, moral, and legal standards.

C. Attitudes:

1. Be aware of the ethical considerations when using ChatGPT and the need of conducting more research on related topic.

Topic	*Task	Time (minutes in class / pre-lesson / after lesson)	Learning Resources
5.1 The Role of AI in Reaching the Sustainable Development Goals		7 mins	Online resources
5.2 AI in Everyday Life	1 & 2	15 mins	Online resources
5.3 AI Advancements: Revolutionizing Professional Fields		5 mins	Online resources
5.4 AI and Global Society		8 mins	Video and articles

*Remarks

- Teachers can assign tasks flexibly (pre-lesson, in-class or assessment), and not all tasks are required to be covered during lesson time.
- Prior to playing any reference video in a unit, it is important for teachers to carefully review the content to ensure its suitability for the teaching purposes intended before proceeding with playing it.

Unit 5 – Societal Impact of AI (II)

5.1 The Role of AI in Reaching the Sustainable Development Goals

The societal impact of AI is vast and far-reaching, with the potential to transform various domains of human life. The objective of utilising **AI for Social Good** is to address significant societal challenges, particularly those affecting **healthcare**, **the environment**, and **our society**. An interesting example of how AI can be applied to promote social good is the "TrailGuard Artificial Intelligence camera" (please see Booklet 2 Unit 7). This project attempts to stop poachers and protect parks and communities.

Recently, United Nations Educational, Scientific and Cultural Organization (UNESCO) released their report on AI for **Sustainable Development Goals** (**SDGs**), grouped into three dimensions: **society, economy, and environment**.

The potential use of AI-based solutions will help us achieve objectives towards **SDGs**. The anticipated results of using AI (numbers in green and orange bars) are shown in the figure below.

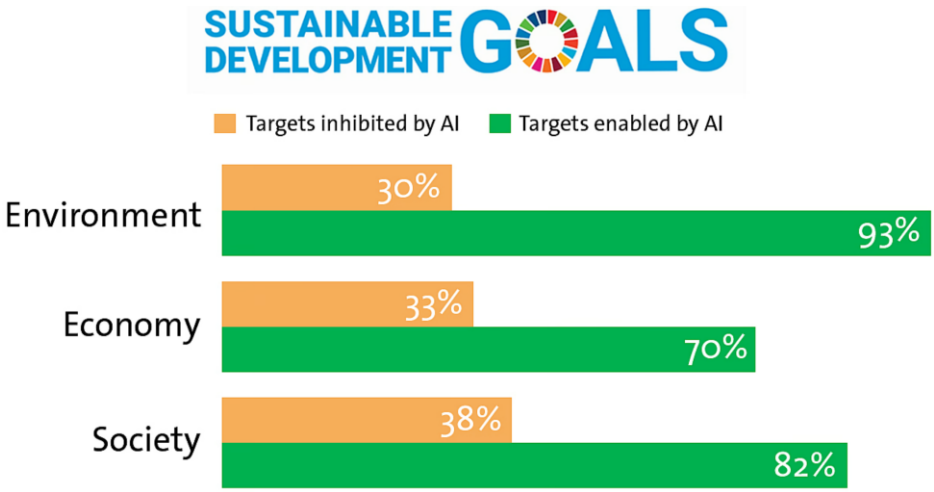


Figure 5.1: Impact of AI in achieving targets from the SDGs

Example: An AI application towards SDGs

“Project Connect” is an initiative by the United Nations Children’s Fund (UNICEF) that employs AI to create a global map of schools and their connectivity using satellite images. This map serves as a valuable tool for identifying areas with inadequate connectivity and informing targeted interventions to address this issue. This initiative fosters the improvement of educational management systems and internet services across all schools, leading to a global advancement in the quality of education. The project aligns with the **SDGs**, targeting the objectives to promote lifelong learning and provide quality education for all.



Figure 5.2: Using AI to map every school in the world

5.2 AI in Everyday Life

AI has become more sophisticated and capable of performing tasks that were previously thought to be exclusive to human intelligence. As a result, AI is becoming an integral part of our daily lives, transforming the way people interact with technology, and creating new possibilities for innovation and advancement.

Task 1: AI-based Application and Its Societal Benefits

Are you aware of any AI-based applications that improves the quality of our everyday life?

Example: “MyObservatory” application from the Hong Kong Observatory.
Hong Kong observatory delivers location-specific weather data, forecasts, warnings and notifications directly to users via “MyObservatory” smartphone application. The Observatory employs AI technology in its weather forecasting and warning services.



Figure 5.3: “MyObservatory” application

Can you describe its societal benefits?

Task 2: Use of Generative AI for Education

Generative AI models such as ChatGPT (Please see Booklet 3 Unit 4) can be valuable tools in education, providing a large volume of examples and content for students to learn from, as illustrated in the example (Figure 5.4). Furthermore, teachers can use generative AI tools to create customized writing prompts and assignments.

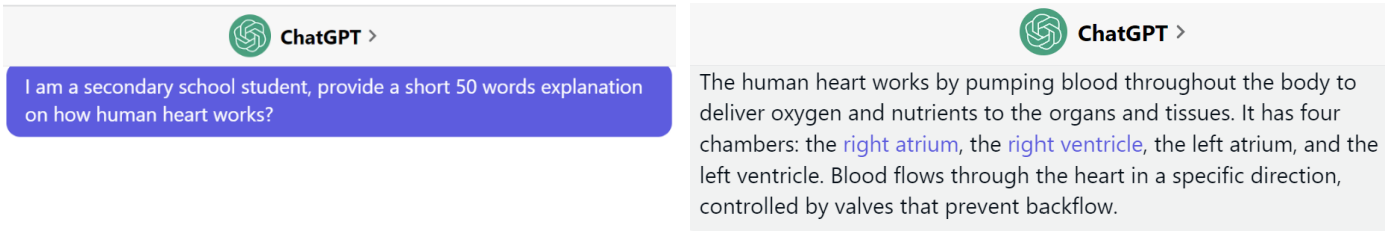


Figure 5.4: Using ChatGPT to ask questions regarding academic subjects

Could you explore ChatGPT and provide a list of how it benefits both teachers and students?

Applications of ChatGPT in education: for teachers	Applications of ChatGPT in education: for students

Despite the potential benefits of using ChatGPT in education, there are concerns about the quality, accuracy, and consistency of AI-generated contents. The content generated by AI may not always be accurate or make sense, and could also repeat myths and stereotypes. Therefore, human validation and oversight are needed before relying on it. While ChatGPT can be used by students for writing assignments or essays, its implementation raises plagiarism and academic integrity issues. As the technology cannot be held accountable for the content it generates, the ethical concerns surrounding this technology highlight the pressing need for further research and development in this field.

5.3 AI Advancements: Revolutionizing Professional Fields

In March 2023, the release of GPT-4, a new generative AI model, showcased its ability to accurately answer questions on US medical licensing exam with a success rate exceeding 90%. Additionally, GPT-4 achieved a score in the top 10% of test takers on the bar exam. The implications of this breakthrough are far-reaching and could potentially change the way these professions are approached, with AI models like GPT-4 providing valuable support and assistance. Moreover, the accessibility of expert knowledge through AI could potentially narrow the gap between underserved communities and quality professional services.

5.4 AI and Global Society

AI is revolutionizing many sectors of society, including healthcare, finance, education, and transportation. While it has the potential to transform our lives, it also presents significant challenges and risks that must be addressed by the global society. AI technologies may not be inherently good or bad, but they still pose a considerable risk. The most probable scenarios for such risks include when AI is programmed with a specific goal that leads to unintended destructive or catastrophic consequences, or when it is intentionally designed to be challenging to turn off, putting human control at risk.

AI has already proven to be a valuable tool that is widely used by people globally. Overall, it is crucial to take into account the societal effects of AI and implement measures to ensure that its development aligns with ethical, moral, and legal standards. With the guidance of the “Asilomar AI Principles”, AI could keep getting better and help more people in the future. The following are the key principles that aim to ensure equitable sharing of AI benefits and responsible development and deployment:

- Implement robust safety and control measures to ensure AI is used safely
- Adhere to universal AI ethical principles
- Carefully consider the impact of AI on society

Learning Summary

- Using AI for social good involves solving important challenges that have an impact on our society, economy, and environment.
- Generative AI models like ChatGPT can be useful for education.
- ChatGPT can be helpful for writing assignments, but it raises concerns about plagiarism and academic integrity, and more research is needed to address these ethical issues.
- The Asilomar AI Principles stress the need to think about how AI will affect society and make sure it is aligned with ethical, moral, and legal standards.

Unit 5 – Assessment Exercise

A. Fill in the blanks

1. ChatGPT can be a valuable tool for students to generate ideas and improve their writing skills, but its use raises concerns about _____ and _____.
2. "Myobservatory" smartphone application contributes to _____ by issuing warnings for extreme weather.
3. _____ is a special project by UNICEF that uses AI to make a map of all _____ in the world in order to promote lifelong learning and to provide quality education for all.

B. Multiple-choice questions

1. Which of the following AI technologies is NOT for social good?

   A. Virtual Assistant
   B. Voice recognition
   C. Autonomous weapons
   D. Smart lamp post

2. Which of the following statements is accurate regarding AI technology?

   A. AI technology is always good and has no risks
   B. AI technology is only risky if it is designed to be easy to turn off
   C. AI technology is inherently bad and should not be used in society.
   D. AI technology can pose risks when it is programmed with a specific goal that leads to unintended consequences.

3. What of the following is an ethical concern associated with the use of autonomous weapons?

   A. Increasing efficiency in military operation
   B. Reducing risk of harm to human soldiers and civilians
   C. Delegating decisions about life and death to machines
   D. Increasing speed and accuracy in target identification

C. Short questions

1. What are the three dimensions of SDGs?

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2. What are the possible concerns associated with using ChatGPT in education?
